

FORM TP 2005201



TEST CODE **02220020**

MAY/JUNE 2005

CARIBBEAN EXAMINATIONS COUNCIL
ADVANCED PROFICIENCY EXAMINATION

ENVIRONMENTAL SCIENCE

UNIT 2: AGRICULTURE, ENERGY AND ENVIRONMENTAL POLLUTION

PAPER 02

2 hours 30 minutes

09 MAY 2005 (p.m.)

READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. **DO NOT** open this examination paper until instructed to do so.
2. This paper consists of **NINE** questions, **THREE** from each Module.
3. Candidates **MUST** answer **SIX** questions, **TWO** from **EACH** Module.
4. Write your answers in the answer booklet provided.

MODULE 1

Answer ANY TWO questions.

1. (a) Outline ONE environmental problem resulting from agricultural practices in a NAMED Caribbean country. [5 marks]
- (b) Based on your response to (a) above, explain the importance of agricultural sustainability in Caribbean societies. [5 marks]
- (c) Describe TWO environmentally sustainable agricultural practices. Make clear in your description why EACH is considered sustainable. [10 marks]

Total 20 marks

2. (a) Describe THREE technological applications in agricultural systems in the Caribbean. [12 marks]
- (b) Discuss TWO positive and TWO negative impacts resulting from the use of technological applications in agricultural systems. [8 marks]

Total 20 marks

3. (a) Distinguish between 'chemical pest control' and 'biological pest control'. [4 marks]
- (b) Discuss ONE advantage and ONE disadvantage of chemical pest control. [8 marks]
- (c) Explain the concept of Integrated Pest Management and why you would recommend its adoption by your country. [8 marks]

Total 20 marks

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MODULE 2

Answer ANY TWO questions.

4. Figure 1 shows the relationship between energy demand and energy generated over the last ten years in a certain Caribbean country. The energy is generated exclusively from steam powered turbines using fossil fuels.

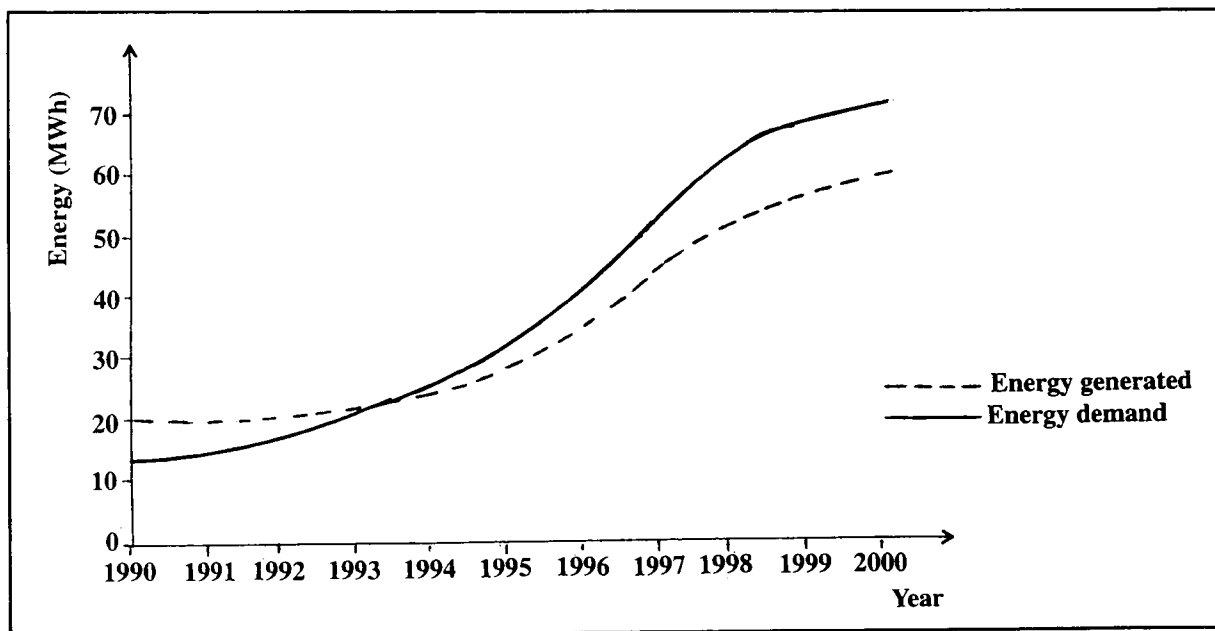


Figure 1. Energy demand and energy generated

- (a) State THREE inferences that can be drawn from the graph. [3 marks]
- (b) Discuss TWO constraints faced by non-oil producing Caribbean countries in satisfying the energy demands of their population. [8 marks]
- (c) Cogeneration is suggested as one approach to address the changing energy demand depicted in Figure 1.
- (i) Explain the process of cogeneration. [4 marks]
- (ii) Do you agree with the suggestion in (c) above? Justify your answer. [5 marks]

Total 20 marks

5. (a) Distinguish between 'nuclear fission' and 'nuclear fusion'. [2 marks]
- (b) Write the equation for the nuclear fission process used in nuclear reactors. [4 marks]
- (c) Explain how electrical energy is generated in a nuclear reactor. [5 marks]
- (d) Discuss THREE issues that could be of concern to a Caribbean country that is considering the use of nuclear fission for generating electricity. [9 marks]

Total 20 marks

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6. (a) Distinguish between 'energy conservation' and 'energy efficiency.' [4 marks]
- (b) Explain the concept of energy-efficient buildings. [4 marks]
- (c) For EACH of the following, state FOUR ways in which the concept of energy-efficient buildings can be applied in the Caribbean.
- (i) Lighting [4 marks]
- (ii) Maintaining comfortable indoor temperature [4 marks]
- (d) Explain how any ONE of your normal activities can be modified to conserve energy. [4 marks]

Total 20 marks

MODULE 3

Answer ANY TWO questions.

7. Figure 2 shows emissions from a number of factories when in operation.

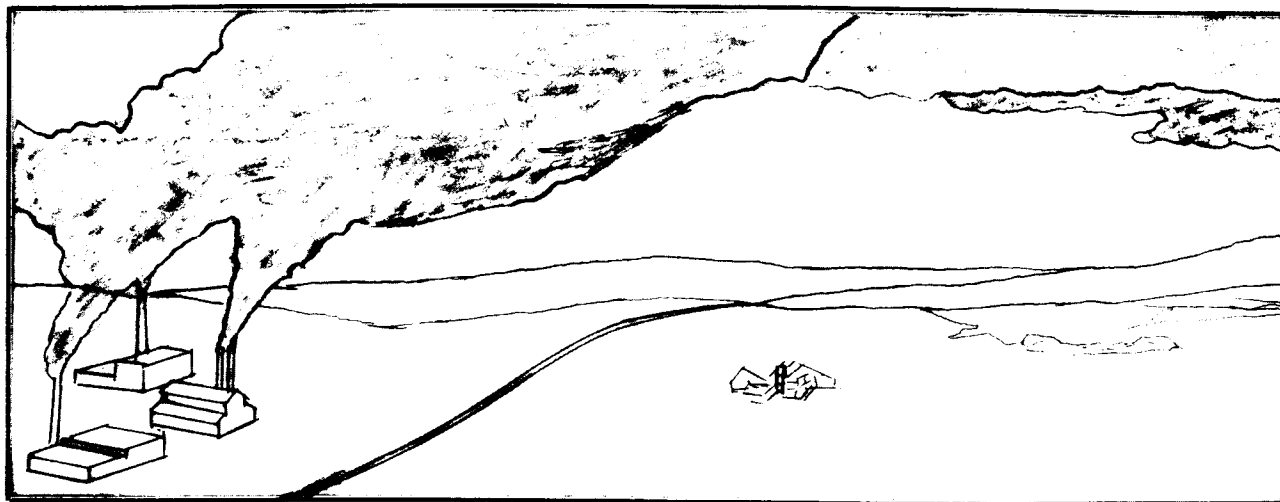


Figure 2. Emissions from a factory

- (a) Suggest TWO pollutants other than volatile organic compounds that may be emitted from the stack. [2 marks]
- (b) State FOUR environmental receptors of ANY pollutant indicated in (a) above. [4 marks]
- (c) Outline the impact of ANY ONE of the pollutants identified in (a) on ONE of the environmental receptors identified in (b) above. [5 marks]
- (d) Describe how scrubbers and electrostatic precipitators can decrease the quantity of emissions from the factory. [9 marks]

Total 20 marks

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8. (a) Explain what is meant by the term 'hazardous waste'. In your explanation, identify ONE example and the source of this type of waste in the Caribbean. [4 marks]
- (b) Define THREE categories of hazardous waste in terms of the impact of EACH on human beings. [6 marks]
- (c) Outline FOUR considerations when planning for the disposal of hazardous waste. [4 marks]
- (d) Describe an appropriate method for the disposal of hazardous waste in a Caribbean country. [6 marks]

Total 20 marks

9. (a) What are 'pathogens'? Include ONE example of a pathogen in your response. [3 marks]
- (b) Identify an agricultural source of pathogens and outline its environmental pathway from this source to water bodies. [3 marks]
- (c) Outline a wastewater treatment process that removes pathogens and nutrients. [11 marks]
- (d) (i) What are 'effluent discharge regulations'? [2 marks]
- (ii) State how they protect water quality. [1 mark]

Total 20 marks

END OF TEST